

Building Blocks Transcript

Now, let us look at the simple building blocks. Let us try to plot a figure. First of all, in any case, when you are plotting a figure, you should have a clarity on what is the x and y axis and what is that you are going to plot.

So, in this case, what we are doing, we are taking numbers 1 to 50 and simply plotting a square function or a parabola y is equal to x square. This is the definition where x is 1 to 50 and y is nothing but power of x raised to for x in x . For every x in x , you are getting y is equal to x square. So, this is the function that we are trying to plot and for plotting, you have to import the figure function.

So, like in math plotlib, whenever you are plotting, you are first creating a figure function, so that the figure area is displayed. Now, likewise, you have in the bokeh environment, you have to first import the figure and then impose the function on that. That is how it works.

So, let us see the command step by step and then take it from there. So, I have imported the necessary code. Now, from bokeh plotting, this is the library bokeh plotting within the bokeh system.

I am importing the figure comma show. I also take a need to import the show function, so that we will once the graphic is done, you have to use the show function to display the figure. So, first I generated the variables, which I already explained.

Then, I am also creating a variable called figure, which is function of a figure. So, I am saying p is p takes the value of a figure and I am plotting a line. I am giving x and y and legend.

I am saying legend is equal to temperature. This is what I have. I just said t e m p legend label and I want to show the results of p . So, what is happening? The figure function is assigned to the variable p and therefore, you are using the line function to draw a line plot.

So, this is how you get. So, this is very simple and similar to what you have done in matplotlib. So, but the most interesting part comes from the interactivity and the nature of the graph that is there.

So, you have got a legend, which is nearby and you have a zoom function here. You can move the graphic here and you can download this as a png or else you can refresh this. So, this is a zoom function.

You can use this for interactivity and you have localised zoom. So, you can do all that and finally, you can reset the graph. This is the interactivity feature that you have in the bokeh.

So, now this becomes a html object in your visualisation space, which allows the audience to interact. If they want to scan on zoom into a particular area, they will be able to do so. So, these are given here.

What are the different functions? So, now we next move to something called glyphs. What are glyphs? Glyph is a visual representation of data. In the example above, the line representing x and y data line is a glyph.

So, this is a bokeh language. The line that we had plotted is called a glyph because you can customise it features or make it interactive. Bokeh supports many different kinds of glyph to be used inside the plots.

We have scatters, wedges, bars, tiles so on. So, let us see an interactive example for adding multiple glyphs. So, again we are first preparing the data x and y values.

I am simply giving the x and y values here. So, which is 1, 2, 3, 4 and we have 2, 5. So, these are x and y values. I have given them.

Then what am I doing? I am adding. First, I have to define the figure. Then P is equal to figure title adding multiple glyphs.

I have done that. Then I am adding P dot line. So, line x values, y values, line width is equal to 2, line label is equal to legend is equal to line.

So, this gives a glyph object with so and so id. Now, I have to show this. Otherwise, it will not be shown.

The inline execution that you have seen in matplotlib is not available here. You have to use the show function. That is how you have got this.

So, what is it doing? It is first creating a variable, calling the figure function and applying the line graph and finally, once it has created a glyph which is stored and now we have to show the figure. Then your glyph is blocked with the same interactivity features. Now, what you can do is like you can add certain additional features to this glyphs which are also additional glyphs in the bokeh environment.

So, what are the additional glyphs that we are trying to do? This plane line is already there and removing it. So, I want to add a circle. Uncomment the line below to add a circle markers on the top of the line.

So, we want to add circle markers to the points at the leg. These four points I want to add circle. So, what I am doing? I am just saying `p.circle` at the same x and y values, size is equal to 10, colour is equal to red and I want to have the legend label called circles and just running the focused selection.

Then again I have to run the show. So, now I am getting the red dots at each place. Now, I want to add square, but at a different place than the y values, I can directly pass the y values or x values, plot the square at these particular values which are different from y values.

For 1, you have 2, you have 1, 5 is equal to 4. You are just shifting it by one digit, so that it is one step away from the current line that you have. So, again I am doing the same. I am running the selection and again I have to do the show.

You can see you have got the square values below the graph and finally, I am doing the legend label. So, we have got the legend also. So, the legend is clickable which means if you click on this that is hidden.

Then if you click on circles, it is circles are hidden. You want to hide it. So, this is what you want to do.

So, you can hide the legend by click legend dot tick dot policy function and is equal to hide will allow you to do this interactivity that you do not want to see circle. You want to see only line. You can just click on those and you can get this interactivity.

Imagine how your graphics interactivity would increase if you start using bokeh and you run it. Now, let us talk about widget. Widgets are small tools that are placed alongside the visualisations to enhance the interactivity features.

Especially, you can work on the widgets. If you are working on time series, you want to give a date range scalar. It would be very useful to give a date range so that people can interact on the date ranges and see the phenomenon for the selected range of dates.

Let us see how does this work. So, you have the date range slider function. You can see the function from bokeh models.

You are importing it and then you are giving date. Date range slider is equal to your setting. This is a variable name, but this is the function date range slider.

You are giving values, first value to the last value, start value and end value. So, these are the values that are there in the slider, the entire range, but you are starting and ending it with different numbers. So, and then simply run this.

So, you got this. So, value is starting from October 1st, 2022, 31st December, 2022. Then starting it is showing 1st July and ending is showing 31st March, 2023.

So, and you can interact this with another graph. For example, you had a time series where the data runs for the entire financial year. You want to do it for each quarter or you can sequentially see for number of days people are attending school or college.

You can vary the range slider and see the interactivity. So, this kind of widgets can also be embedded into your graphics and make your graphics more interactive. So, this is an audience narrative control feature where they want to do more analytics on the features that you have presented in the graph.

They can simply do this by themselves and draw the conclusions they want to draw from the data. Now, let us see the layouts. What are layouts? Layouts are preset design formats that are available for people to explore and generate standard graphics.

So, a Bokeh document can consist of a single plot or it can have a multiple plot and you use the layout function to arrange them into a single plot and bring more emphasis in a consistent manner. Now, let us see what is the use of layouts. So, from `Bokeh.layouts`, I am importing `row` and `Bokeh.plotting`, I am importing `figure` and show which already we had done.

So, I am importing `row` function and then `x` is nothing but 1 to 11. Then `y0` is equal to `x`, `y1` is equal to `10 minus i` for `i` in `x`, which means `10 minus 11`, `9`, like I mean `10 minus 10`, `10 minus 9`, `10 minus`. So, you will have from `10` to `0`. Then you have absolute of `5 minus 5` for `i` in `x`. You have three different functions.

Here we have defined them. Then we are trying to plot each of these variables. We want to plot `x` with `y0`, `x` with `y1` and `x` with `y2` using the scatter plot and each case we have to define this as a variable using the `figure` function and then you have said height and width of the graphic, `250` is to `250` and you want to fill the background with particular colours and you want to also distinguish them with different marker and the marker size or colour.

So, you have scatter 1, you have scatter 2 and you have scatter 3 and finally you want to show them as one or three variables in a particular row. The graphs are here, the variables are the figures are the variables. In one row, you want to show `s1`, `s2`, `s3`.

So, this is what the function will do and we are just running this. So, you got 1, 2, 3. So, once we run this, you get three nice graphics arranged in the same row with interactivity features. So, it is 0 to 10.



So, simply y is equal to x . Then it is inverse of that 10 to 0. Then you have a B shape modulus function which has a value of 0 at 5. So, now I want to plot instead of rows, I want to plot in column. So, I am just writing `import column`. So, I am doing this.

So, I am doing this. So, I am getting it in column. Now, imagine how many graphics you can arrange in a nice template or layout format using the bokeh and still provide the interactivity for each of this.

So, we have used the facets feature in the, facets feature in the Matplotlib and in the Seaborn library. You can improvise that with the bokeh environment. Thank you.

-----End-----